

Bachelor's thesis

Hidden Markov Models for High Frequency FX Pairs Trading

Regime-gated co-integration strategies on currency tick data

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Abstract

This thesis asks whether Hidden Markov Models (HMMs) add value to a classical pairs-trading rule on high-frequency currency tick data (Gatev et al. 2006; Vidyamurthy 2004; Avellaneda and Lee 2010; Krauss 2017). Bid-ask tick data from Dukascopy Bank SA (2026) for three currency-pair constructions—AUD/USD–NZD/USD, EUR/NOK–EUR/SEK, and GBP/USD–EUR/USD—are pre-averaged on number of ticks (Jacod et al. 2017) and studied across three independent monthly episodes (August 2024, September 2024, and August 2025), chosen for active price movement under structurally different macro backdrops. The pre-averaged mid-price series is used to build a rolling cointegration spread (Engle and Granger 1987; Dickey and Fuller 1979). The textbook z -score rule on this spread is the baseline strategy; a Markov-switching AR(1) on the same spread (Hamilton 1989; Krolzig 1997) is the regime model. The HMM is used as a regime gate *on top of* the baseline, not as an independent forecaster: an AR-gated variant switches the baseline off in a high-volatility drifting regime, and an MS-AR variant widens the entry threshold continuously with the drifting-regime probability. All strategies are evaluated under a walk-forward back-test with realistic transaction costs (Roll 1984; López de Prado 2018). We find that the HMM functions as an interpretable defensive filter rather than as a source of additional risk-adjusted return: it correctly identifies high-variance episodes of the spread, but on the pairs where the baseline is already profitable, removing exposure during those episodes removes profitable exposure as well, and risk-adjusted ratios fall accordingly (Sharpe 1994; Sortino and Meer 1991).

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1 Introduction

1.1 The quantitative revolution

Modern financial markets are mathematical objects. The trading pits, voice execution, and discretionary judgement of the past have been replaced by electronic limit-order books, automated market makers, statistical arbitrage engines, and latency-sensitive execution algorithms (Lewis 2014; Aldridge 2013; Biais et al. 2015). Prices are no longer just economic summaries of fundamentals; they are also high-dimensional data streams generated by the interaction of agents, algorithms, inventories, order flows, and information shocks (O’Hara 1995; Hasbrouck 2007; Hautsch 2012).

From a mathematical perspective, the deeper issue is not speed but *non-stationarity*: financial time series are noisy, adaptive, and adversarial, so any exploitable structure is weak, transient, and rapidly arbitrated away (Cont 2001). The trader is no longer simply asking whether an asset is “cheap” or “expensive”, but whether a small statistical regularity can be detected, validated, traded, and risk-managed after costs (López de Prado 2018; Aït-Sahalia and Jacod 2014). Jim Simons is a useful symbol of this view: a mathematician who treated finance as a signal-processing problem rather than as economic forecasting (Zuckerman 2019).

1.2 FX microstructure and pairs trading

The foreign exchange (FX) market is one of the most natural settings in which to study this problem: it is globally traded, electronically intermediated, and active nearly twenty-four hours a day. At low frequencies, exchange rates are well approximated by continuous diffusions (Bachelier 1900; Black and Scholes 1973; Merton 1973). At the tick level, however, prices move in small discrete increments, quotes update asynchronously, and the price path becomes a sequence of small jumps and microstructural frictions (Roll 1984; Jacod et al. 2017). A single stationary Gaussian model does not capture this picture.

Currency pairs with shared economic or geographic exposure are natural candidates for *pairs trading*: rather than betting on the level of one exchange rate, one bets on temporary deviations between two related ones reverting to a common stochastic trend (Gatev et al. 2006; Vidyamurthy 2004; Avellaneda and Lee 2010; Krauss 2017). The three constructions used here—AUD/USD vs NZD/USD, EUR/NOK vs EUR/SEK, and GBP/USD vs EUR/USD—span a range of liquidity and dependence regimes.

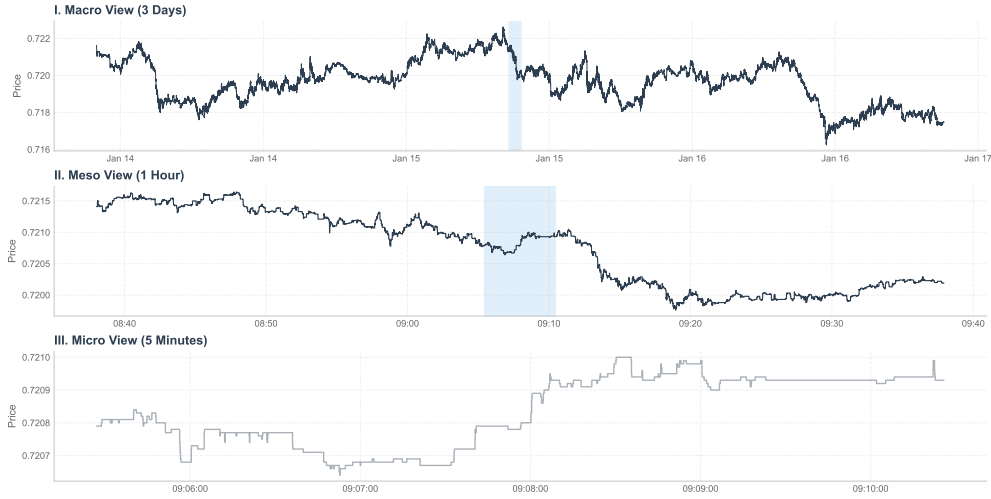


Figure 1: FX tick data at three scales. Top: daily horizon, approximately diffusive. Middle: hourly horizon, localised mean reversion. Bottom: tick level, discrete and microstructural.

1.3 Non-stationarity and regimes

The main statistical difficulty is non-stationarity: volatility clusters, correlations rise and fall, and liquidity varies intraday (Engle 1982; Bollerslev 1986; Cont 2001). A trading rule that works in one regime may fail in another. This motivates the language of *market regimes*: persistent latent states, each with its own statistical properties, switching according to some stochastic mechanism (Hamilton 1989).

We observe only the emitted data: returns, spreads, and trading outcomes. The natural modelling tool is therefore a Hidden Markov Model (HMM), which assumes that an observable process is driven by an unobserved Markov chain whose state determines the distribution of the observation (Rabiner 1989; Cappé et al. 2005; Krolzig 1997).

1.4 The role of the HMM in this thesis

The exercise is deliberately layered.

1. We build a single tradable signal: the rolling cointegration spread S_t (Engle and Granger 1987; Dickey and Fuller 1979), standardised into a z -score Z_t . The *baseline* pairs-trading rule enters and exits positions on Z_t alone (Gatev et al. 2006; Vidyamurthy 2004).
2. We back-test the baseline first under realistic transaction costs. This establishes the benchmark against which any added complexity must justify itself.
3. We then introduce a Markov-switching AR(1) on S_t (Hamilton 1989; Krolzig 1997) to recover a hidden chain $K_t \in \{1, \dots, K\}$ representing the active regime.
4. The HMM is used *as a regime gate on top of the baseline signal*, not as an independent forecaster. The trading direction stays dictated by Z_t ; the HMM’s role is to suppress or down-weight trading during high-volatility, weak-mean-reversion regimes. In short: *the baseline gives the signal, the HMM decides when not to trade*.

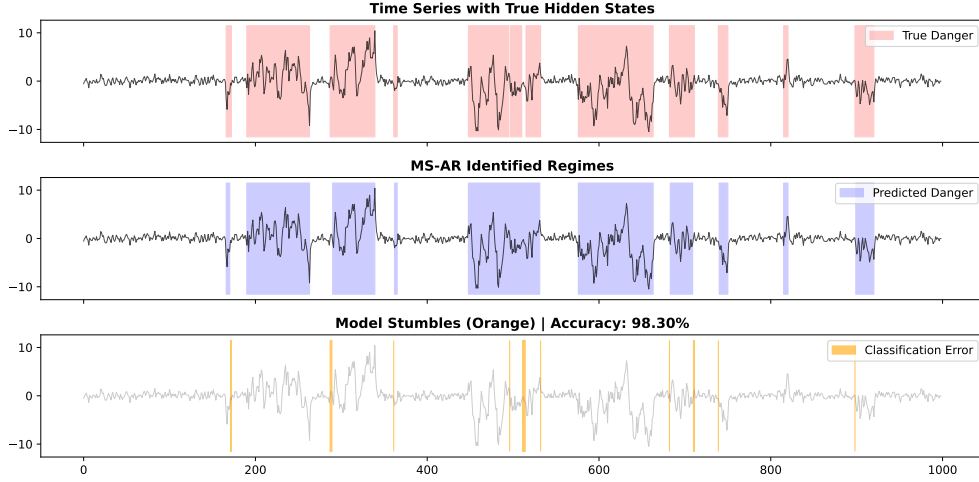


Figure 2: HMM regime recovery on a simulated two-regime AR(1). Top: simulated series with the true regime shaded. Middle: HMM-inferred danger regime. Bottom: classification error.

Because the regime-aware strategies inherit the baseline’s direction, any difference between them and the baseline is attributable to the gate alone—which is why baseline results are reported *before* regime-model results.

1.5 Structure

Section 2 describes the data. Section 3 develops the cointegration spread and the baseline rule. Section 4 introduces the AR/HMM machinery and the regime-aware strategies. Section 5 specifies the walk-forward back-test and the performance metrics. Section 6 reports the empirical results. Section 7 concludes.

2 Data and pre-processing

We use high-frequency bid–ask FX tick data from Dukascopy Bank SA (2026), loaded into Python with `pandas` (McKinney 2010) and `NumPy` (Harris et al. 2020). The full

data-processing and modelling pipeline, including the back-test driver and the figures reproduced in this thesis, is released as the open-source repository `stk-mat2011` (Furnes 2026). The two legs of a pair are indexed by $i \in \{A, B\}$. The three pair constructions are AUD/USD–NZD/USD, EUR/NOK–EUR/SEK, and GBP/USD–EUR/USD; they span a deliberate range of liquidity, transaction costs, and pair-dependence (Gatev et al. 2006; Krauss 2017).

Rather than running a single continuous back-test, we study three independent one-month sub-samples: *August 2024*, *September 2024*, and *August 2025*. These months were chosen because they cover periods of active price movement on all three pairs while spanning distinct macroeconomic backdrops, which gives three structurally different episodes on which the same machinery can be exercised. Reporting the three episodes side by side is more informative than aggregating them: each month is treated as a self-contained case study, and any difference in conclusions across months is itself a result.

Bid and ask streams are sorted by timestamp and aligned by a backward as-of merge (Hasbrouck 2007). We write $P_t^{\text{bid},i}$ and $P_t^{\text{ask},i}$ for the prevailing bid and ask of leg i at time t . The mid-price and basis-point half-spread are

$$P_t^{\text{mid},i} = \frac{1}{2}(P_t^{\text{bid},i} + P_t^{\text{ask},i}), \quad \text{HS}_t^i = \frac{1}{2} \frac{P_t^{\text{ask},i} - P_t^{\text{bid},i}}{P_t^{\text{mid},i}} \times 10^4. \quad (1)$$

HS_t^i approximates the cost of crossing the market for one leg (Roll 1984). Since pairs trading enters and exits both legs simultaneously, the combined spread cost is central to the back-test.

2.1 Pre-averaging on number of ticks

Tick data are dominated by microstructure noise: bid–ask bounce, discreteness, and asynchronous quote updates contaminate the observed price relative to the latent efficient price (Roll 1984; Hasbrouck 2007; Hautsch 2012). Following Jacod et al. (2017), we therefore apply a *pre-averaging* step in tick time: each block of L consecutive ticks is collapsed into a single observation by averaging the mid-prices within the block. With $\{P_{t_n}^{\text{mid},i}\}_n$ denoting the synchronised tick-time mid-price for leg i , the pre-averaged series at block index k is

$$\tilde{P}_k^i = \frac{1}{L} \sum_{n=(k-1)L+1}^{kL} P_{t_n}^{\text{mid},i}, \quad k = 1, 2, \dots \quad (2)$$

The block size L is the same for both legs of a pair, so the pre-averaged legs remain perfectly synchronised. Pre-averaging is preferred here to fixed calendar-time sampling because it produces an observation index that is more nearly homogeneous in microstructure activity and substantially reduces the noise variance of the resulting return series (Jacod et al. 2017; Aït-Sahalia and Jacod 2014; Andersen et al. 2000). All subsequent quantities—log returns, cointegration spread, z -score, and HMM observations—are computed on the pre-averaged series $\tilde{P}_k^i$. For notational simplicity we continue to write $P_t^{\text{mid},i}$ for the price series fed to the models, with the convention that t now indexes pre-averaged blocks rather than individual ticks.

3 Pairs trading with the baseline

This section defines the baseline strategy as a self-contained pairs-trading rule (Gatev et al. 2006; Vidyamurthy 2004). The regime-aware extensions of Section 4 keep this rule's entry direction and only modify when it is allowed to trade.

3.1 Cointegration spread

Two non-stationary log-price series $\log P_t^{\text{mid}, A}$ and $\log P_t^{\text{mid}, B}$ are cointegrated if some linear combination is stationary (Engle and Granger 1987):

$$S_t = \log P_t^{\text{mid}, A} - \beta \log P_t^{\text{mid}, B}. \quad (3)$$

The hedge ratio β is estimated by the Engle–Granger two-step procedure: OLS regression of $\log P^{\text{mid}, A}$ on $\log P^{\text{mid}, B}$, with stationarity of the residuals verified by an Augmented Dickey–Fuller test (Dickey and Fuller 1979; Said and Dickey 1984). High-frequency FX is not globally cointegrated: β drifts with liquidity, session structure, and pair-specific shocks. We therefore estimate β_t by rolling OLS with window W_β . Writing $x_s = \log P_s^{\text{mid}, B}$, $y_s = \log P_s^{\text{mid}, A}$, $\mathcal{W}_t = \{t - W_\beta + 1, \dots, t\}$,

$$\beta_t = \frac{W_\beta \sum_{s \in \mathcal{W}_t} x_s y_s - (\sum_{s \in \mathcal{W}_t} x_s)(\sum_{s \in \mathcal{W}_t} y_s)}{W_\beta \sum_{s \in \mathcal{W}_t} x_s^2 - (\sum_{s \in \mathcal{W}_t} x_s)^2}, \quad \alpha_t = \frac{1}{W_\beta} \left(\sum_{s \in \mathcal{W}_t} y_s - \beta_t \sum_{s \in \mathcal{W}_t} x_s \right). \quad (4)$$

The cointegration spread and the associated spread return are

$$S_t = \log P_t^{\text{mid}, A} - \beta_t \log P_t^{\text{mid}, B} - \alpha_t, \quad \Delta S_t = r_t^A - \beta_{t-1} r_t^B, \quad (5)$$

with $r_t^i = \log P_t^{\text{mid}, i} - \log P_{t-1}^{\text{mid}, i}$. The lag β_{t-1} ensures the hedge ratio is observable when the return is realised.

3.2 The rolling z -score and the baseline rule

The baseline reads S_t through a rolling z -score over a window of length W_z :

$$Z_t = \frac{S_t - \mu_t^S}{\sigma_t^S}, \quad \mu_t^S = \frac{1}{W_z} \sum_{s=t-W_z+1}^t S_s, \quad (\sigma_t^S)^2 = \frac{1}{W_z} \sum_{s=t-W_z+1}^t (S_s - \mu_t^S)^2. \quad (6)$$

With entry threshold z_q and exit threshold z_x , the position $\pi_t \in \{-1, 0, +1\}$ is updated by

$$\pi_t = \begin{cases} +1 & \pi_{t-1} = 0, Z_t < -z_q, \\ -1 & \pi_{t-1} = 0, Z_t > +z_q, \\ 0 & \pi_{t-1} \neq 0, |Z_t| \leq z_x, \\ \pi_{t-1} & \text{otherwise.} \end{cases} \quad (7)$$

A long-spread position is opened when S_t is sufficiently below its rolling mean and closed when it returns toward zero; the short rule is symmetric (Gatev et al. 2006). For every unit of leg A we trade $\hat{\beta}$ units of leg B , so the resulting portfolio is exposed to the relative move between the two legs rather than to FX direction (Vidyamurthy 2004).

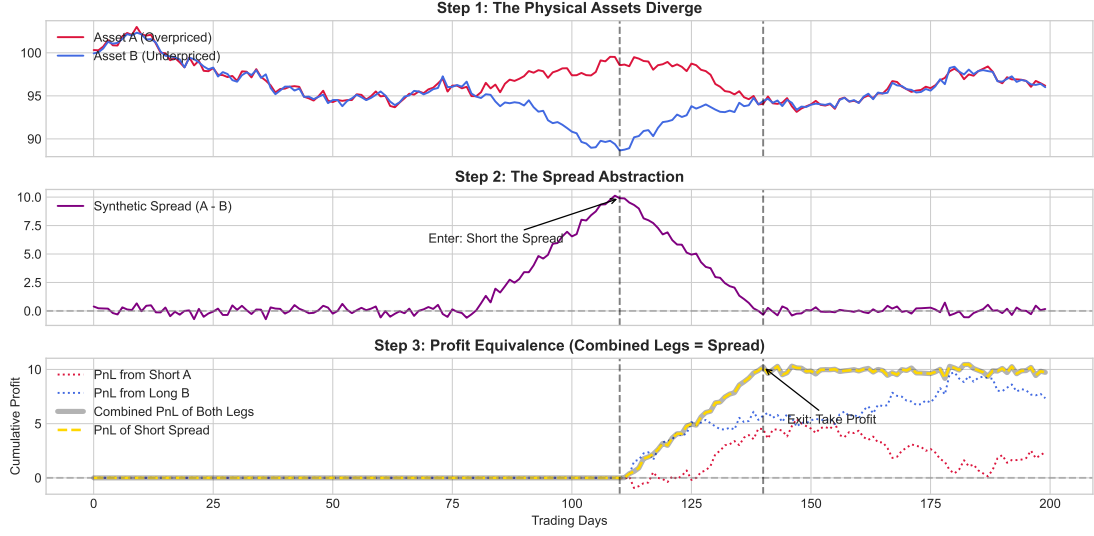


Figure 3: Mechanics of pairs trading. The combined long/short portfolio is exposed only to the relative move between the two legs.

A do-nothing reference $\pi_t^{\text{BH}} = +1$ for every t (Buy & Hold) earns ΔS_t at each step, pays no transaction costs, and provides a passive drift benchmark.

3.3 Why the baseline is not enough

The baseline assumes S_t is governed by a single stationary process, so that fixed (z_q, z_x) correctly characterise “far from equilibrium” for every t . Two empirical features of FX tick data violate this assumption: (i) squared spread returns exhibit strong positive autocorrelation, consistent with conditional heteroskedasticity (Engle 1982; Bollerslev 1986; Cont 2001); and (ii) a spread that mean-reverts quickly in quiet periods may drift directionally for hundreds of bars during news-driven or one-sided-flow episodes (Hasbrouck 2007). The fixed-threshold rule has no mechanism to distinguish these cases. This motivates the regime-aware extensions below.

4 Regime-aware models

We introduce the AR/HMM machinery on a generic series, then specialise to the spread S_t .

4.1 Autoregressive benchmark

The autoregressive model of order p projects a series Y_t on its p most recent past values (Hamilton 1994; Tsay 2010):

$$Y_t = c + \phi_1 Y_{t-1} + \cdots + \phi_p Y_{t-p} + \varepsilon_t, \quad \varepsilon_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2). \quad (8)$$

The simplest non-trivial case is AR(1):

$$Y_t = c + \phi Y_{t-1} + \varepsilon_t, \quad (9)$$

with unconditional mean $c/(1 - \phi)$ for $|\phi| < 1$. The AR(1) captures momentum ($\phi > 0$) or immediate mean reversion ($\phi < 0$), but only if (c, ϕ, σ^2) are constant over the sample. We use (9) as a building block, applied to $Y_t = S_t$ in Section 4.3.

4.2 Hidden Markov chain

To relax the structural-stability assumption we let the parameters of (9) be piecewise constant, switching with an unobserved discrete-time Markov chain $K_t \in \{1, \dots, K\}$ (Hamilton 1989; Rabiner 1989; Krolzig 1997; Cappé et al. 2005). The dynamics of K_t are governed by a $K \times K$ transition matrix $\mathbf{P} = (p_{ij})$ and an initial distribution $\nu = (\nu_1, \dots, \nu_K)^\top$:

$$p_{ij} = \mathbb{P}(K_t = j \mid K_{t-1} = i), \quad \sum_{j=1}^K p_{ij} = 1 \quad \forall i, \quad \nu_k = \mathbb{P}(K_1 = k). \quad (10)$$

The Markov assumption is deliberately simple: the future depends on the past only through the current state. The framework is nonetheless rich enough to capture persistence, sudden switching, and regime-dependent parameters within a single probabilistic model (Rabiner 1989).

4.3 Markov-switching AR(1) on the spread

We specialise to S_t from (5) and take $K = 2$. The joint generative model is

$$S_t \mid S_{t-1}, K_t = k \sim \mathcal{N}(c^{(k)} + \rho^{(k)} S_{t-1}, (\sigma^{(k)})^2), \quad \mathbf{P} = \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix}. \quad (11)$$

This is an AR(1) on the spread with all three parameters $(c^{(k)}, \rho^{(k)}, \sigma^{(k)})$ switching (Hamilton 1989; Krolzig 1997). For $|\rho^{(k)}| < 1$ the regime mean is $\mu^{(k)} = c^{(k)}/(1 - \rho^{(k)})$, yielding the mean-deviation form

$$S_t = \mu^{(k)} + \rho^{(k)}(S_{t-1} - \mu^{(k)}) + \varepsilon_t^{(k)}, \quad \varepsilon_t^{(k)} \sim \mathcal{N}(0, (\sigma^{(k)})^2). \quad (12)$$

Values of $\rho^{(k)}$ close to zero correspond to fast mean reversion; values close to unity, to near-random-walk behaviour. Before estimation, S_t is winsorised at $\text{median}(S) \pm 4 \cdot \text{std}(S)$ and rescaled by 10^4 for numerical stability.

4.4 Estimation by EM (Baum–Welch)

The HMM parameter set is

$$\Theta = \{ \nu, \mathbf{P}, c^{(k)}, \rho^{(k)}, \sigma^{(k)} : k = 1, \dots, K \}, \quad (13)$$

estimated by maximum likelihood with the Expectation–Maximisation algorithm (Dempster et al. 1977), which for HMMs is the Baum–Welch procedure (Baum et al. 1970; Rabiner 1989). The E-step computes the smoothed regime posteriors

$$\gamma_t(k) = \mathbb{P}(K_t = k \mid S_1, \dots, S_T; \Theta) \quad (14)$$

via the forward–backward algorithm. The M-step updates $(c^{(k)}, \rho^{(k)}, \sigma^{(k)})$ and $\mathbf{P}$ to maximise the expected complete-data log-likelihood. The iteration is run to a small tolerance, restarted from several random seeds to mitigate convergence to local maxima (Cappé et al. 2005).

4.5 Model selection

The number of states K and AR order p are chosen by the Bayesian Information Criterion (Schwarz 1978). With m free parameters, T observations, and maximised likelihood $\hat{L}$,

$$\text{BIC} = m \ln T - 2 \ln \hat{L}. \quad (15)$$

The BIC penalises complexity with a term that scales in $\ln T$, which is aggressive in the large samples produced by high-frequency tick data and is asymptotically consistent for the true model when it exists (Hastie et al. 2009; Tsay 2010). It is also the standard choice for hidden Markov order selection (Cappé et al. 2005; Krolzig 1997), which is prone to over-fit when more lenient criteria are used. We sweep over a grid of (K, p) and retain the configuration that minimises BIC for the trading simulations.

4.6 Regime classification

With $K = 2$, regimes are labelled by their innovation variance:

$$k_{\text{MR}} = \arg \min_k \sigma^{(k)}, \quad k_{\text{DR}} = \arg \max_k \sigma^{(k)}, \quad (16)$$

giving a mean-reverting (MR) regime and a drifting (DR) regime. The smoothed posteriors are $\gamma_t^{\text{MR}} = \mathbb{P}(K_t = k_{\text{MR}} \mid S_1, \dots, S_T)$ and $\gamma_t^{\text{DR}} = 1 - \gamma_t^{\text{MR}}$. In live trading, only past observations are available; the filtered posterior $\mathbb{P}(K_t = k_{\text{MR}} \mid S_1, \dots, S_t)$ is then used inside the walk-forward back-test, while the smoothed posterior is reserved for diagnostic figures.

4.7 Regime-aware strategies

We build two regime-aware strategies on top of the baseline (7); both inherit its entry/exit directions and only change *when* the rule is allowed to trade.

AR-gated. A binary gate with danger threshold $\delta \in (0, 1)$,

$$G_t = \mathbb{1}\{\gamma_t^{\text{MR}} \geq 1 - \delta\}, \quad (17)$$

modifies the rule to

$$\pi_t = \begin{cases} 0 & G_t = 0 \text{ (liquidate, stay flat),} \\ \pi_t^{\text{baseline}} & G_t = 1. \end{cases} \quad (18)$$

The gate is binary: trade exactly like the baseline, or sit in cash. It also applies to existing positions: if we are long the spread and the HMM flags danger, we go flat at the next bar.

MS-AR. A continuous, regime-weighted entry threshold,

$$z_q^{\text{eff}}(t) = \gamma_t^{\text{MR}} z_q + \gamma_t^{\text{DR}} z_v, \quad z_v \geq z_q, \quad (19)$$

replaces z_q in (7) by $z_q^{\text{eff}}(t)$. When MR is highly probable, $z_q^{\text{eff}} \approx z_q$ and the strategy trades like the baseline; when DR dominates, a larger $|Z_t|$ deviation is required before entering.

4.8 Strategy summary

Table 1: Strategies. All share the cointegration spread S_t and rolling z -score Z_t ; they differ in whether and how γ_t^{MR} enters the rule.

Strategy	Always long?	Entry threshold	Gate	Reads HMM?
Buy & Hold	Yes	—	—	No
Baseline	No	z_q	1	No
AR	No	z_q	$\mathbb{1}\{\gamma_t^{\text{MR}} \geq 1 - \delta\}$	Yes (binary)
MS-AR	No	$\gamma_t^{\text{MR}} z_q + \gamma_t^{\text{DR}} z_v$	1	Yes (continuous)

5 Back-test protocol and metrics

5.1 Transaction costs and walk-forward evaluation

Every position change in leg i pays the half-spread HS_t^i of (1), plus a fixed commission (Aldridge 2013; López de Prado 2018). The round-trip cost of one pair-position is therefore approximately $\text{HS}^A + \hat{\beta} \text{HS}^B$ in basis points. Strategies are evaluated in a walk-forward structure: parameters Θ and the hyperparameters $(W_\beta, W_z, z_q, z_x, z_v, \delta)$ are estimated on a training segment and applied to a disjoint out-of-sample test segment, with windows then slid forward (López de Prado 2018; Hautsch 2012).

5.2 Performance metrics

Let R_t denote the strategy return at bar t , $\mu_R = \mathbb{E}[R]$, $\sigma_R = \sqrt{\text{Var}(R)}$, V_t the portfolio value with running peak $V_t^* = \max_{s \leq t} V_s$, and R_f a benchmark risk-free rate (set to zero). Statistical metrics describe the shape of the return distribution; financial metrics translate it into capital outcomes (Tsay 2010; Cont 2001).

$$\sigma_{\text{ann}} = \sigma_R \sqrt{T}, \quad \mathcal{S} = \frac{\mathbb{E}[(R - \mu_R)^3]}{\sigma_R^3}, \quad \mathcal{K} = \frac{\mathbb{E}[(R - \mu_R)^4]}{\sigma_R^4}, \quad (20)$$

$$\text{VaR}_\alpha = -Q_{1-\alpha}(R), \quad \text{CVaR}_\alpha = -\mathbb{E}[R \mid R \leq -\text{VaR}_\alpha], \quad \text{TF} = \frac{\text{CVaR}_\alpha}{\text{VaR}_\alpha}, \quad (21)$$

with $\alpha = 0.95$ (Artzner et al. 1999). The drawdown family uses $D_t = (V_t^* - V_t)/V_t^*$:

$$\text{MDD} = \max_t D_t, \quad \text{UI} = \sqrt{\frac{1}{N} \sum_t D_t^2}, \quad (22)$$

where UI is the Ulcer index of Martin and McCann (1989); the maximum drawdown is treated in Magdon-Ismael and Atiya (2004). Risk-adjusted ratios relate excess return to risk:

$$\text{SR} = \frac{\mu_R - R_f}{\sigma_R} \quad (\text{Sharpe 1994}), \quad \text{Sortino} = \frac{\mu_R - R_f}{\sigma_d} \quad (\text{Sortino and Meer 1991}), \quad \text{Calmar} = \frac{\text{AR}}{\text{MDD}}. \quad (23)$$

Trade-level diagnostics (win rate, market exposure, profit factor, payoff ratio) summarise frequency and size; see Section 6 for their interpretation.

6 Empirical results and discussion

This section reports the empirical results. Following the structure of Sections 3–4, we first inspect the data and pair-level diagnostics, then back-test the baseline rule, then add the regime-aware extensions. All numbers are reported net of the transaction-cost assumptions of Section 5.1 (Aldridge 2013; López de Prado 2018).

6.1 Data and pair diagnostics

The three pairs differ substantially in liquidity and round-trip cost (Roll 1984; Hasbrouck 2007): GBP/USD–EUR/USD is the cheapest, AUD/USD–NZD/USD is moderately liquid, and EUR/NOK–EUR/SEK is materially more expensive. All six leg-level return series are strongly non-Gaussian, with high kurtosis and significant Ljung–Box statistics on squared returns, consistent with conditional heteroskedasticity (Engle 1982; Bollerslev 1986; Cont 2001).

Rolling cointegration tests (Engle and Granger 1987; Dickey and Fuller 1979; Said and Dickey 1984) reject the no-cointegration null only intermittently for AUD/USD–NZD/USD and GBP/USD–EUR/USD; EUR/NOK–EUR/SEK rejects globally but has a very long estimated half-life, which makes it economically weak for high-frequency mean reversion (Krauss 2017).

6.2 Baseline back-test performance

We back-test the baseline rule (7) under the walk-forward protocol of Section 5.1. This is the benchmark that any regime-aware extension must beat to justify the added complexity (Gatev et al. 2006; Vidyamurthy 2004).

The baseline separates the three pairs cleanly. AUD/USD–NZD/USD is strongly profitable on a risk-adjusted basis, with a high Sharpe ratio and a small maximum drawdown (Sharpe 1994; Magdon-Ismail and Atiya 2004). GBP/USD–EUR/USD is mildly profitable, with modest Sharpe and Calmar ratios. EUR/NOK–EUR/SEK loses money across the sample: weak rolling cointegration and wide spreads destroy any apparent statistical edge, exactly the scenario Krauss (2017) flag for high-cost statistical-arbitrage pairs.

This is itself a useful first result: the baseline back-test is a tradeability filter. Two of the three pairs pass; one fails, before any regime-aware machinery is applied.

6.3 Regime-aware extensions

We add the AR-gated and MS-AR strategies of Section 4.7 on top of the baseline. Both reuse the baseline’s directional rule and only change when it is allowed to trade, so any difference in performance is attributable to the regime gate alone.

Three findings emerge.

(i) The HMM functions as a defensive filter. Across all three pairs, the regime gate sharply reduces market exposure, the number of trades, and annual volatility. The HMM is doing what Hamilton (1989) and Krolzig (1997) would predict it should do: it identifies high-variance episodes of the spread, and the strategy spends less time in the market during those episodes.

(ii) Defensive behaviour does not translate into higher risk-adjusted return. On the two pairs where the baseline is already profitable, the Sharpe ratio falls when the gate is added: removing the high-variance regime removes profitable exposure as well as dangerous exposure. This is the empirical version of an old observation about regime models: the regime that is bad for variance is not always bad for expected return (Cappé et al. 2005; Rabiner 1989). On GBP/USD–EUR/USD the gate even *increases* the maximum drawdown, by stripping out profitable exposure that would otherwise have offset losing periods.

(iii) On the broken pair, the HMM only limits the damage. On EUR/NOK–EUR/SEK the regime gate reduces the size of the loss and the maximum drawdown but

cannot manufacture a profitable strategy where the underlying microstructural signal does not exist (Krauss 2017).

6.4 Synthesis

Bringing these observations together: the HMM identifies high-risk states correctly, the smoothed posteriors and regime-specific variances behave qualitatively as expected (Rabiner 1989; Cappé et al. 2005), but the identified regimes are not perfectly aligned with profitable opportunities. The MS-AR variant produces a smoothed version of the binary AR gate and does not add value on top.

Posed as in Section 1.4—does the regime gate add value relative to the baseline, or mainly reduce exposure?—the empirical answer is the latter, in this sample and under these costs. The HMM is a valuable diagnostic and a useful defensive device. It is not, on its own, a source of additional risk-adjusted return.

This is a constructive answer rather than a negative one. The HMM provides interpretable, mathematically structured information about the local environment of the spread; what it does not do is automatically convert that information into superior trading performance. A natural direction for future work, discussed in Section 7, is to combine the regime posteriors with a different entry signal (for example, a momentum signal active in the drifting regime, as in Avellaneda and Lee (2010)) rather than only as a gate on a mean-reversion signal.

7 Conclusion

This thesis has tested whether Hidden Markov Models add value to a classical pairs-trading rule on high-frequency FX tick data. The exercise was deliberately layered: a single baseline signal (the rolling z -score of the cointegration spread (Engle and Granger 1987; Gatev et al. 2006; Vidyamurthy 2004)) was first treated as a self-contained strategy and back-tested under realistic transaction costs and a walk-forward protocol (Aldridge 2013; López de Prado 2018); the HMM was then introduced as a regime gate (Hamilton 1989; Rabiner 1989; Krolzig 1997), with two concrete instantiations (a binary AR-gated rule and a probability-weighted MS-AR threshold) that leave the baseline’s directional entry rule unchanged.

The findings are summarised as follows.

1. **The baseline rule separates the three pairs cleanly.** AUD/USD–NZD/USD is strongly profitable, GBP/USD–EUR/USD is mildly profitable, and EUR/NOK–EUR/SEK is unprofitable. The baseline back-test is itself a diagnostic for whether a pair is tradeable under high-frequency frictions (Krauss 2017).
2. **The HMM is a defensive filter, not a return enhancer.** The regime gate sharply reduces market exposure and annual volatility on every pair. On the two profitable pairs, however, it removes profitable exposure as well as dangerous exposure, and risk-adjusted ratios fall (Sharpe 1994; Sortino and Meer 1991). On the unprofitable pair, the HMM only limits the damage.
3. **The MS-AR threshold smoothing does not improve on the binary AR gate** on a risk-adjusted basis.

4. **The model is interpretable.** The smoothed regime posteriors, regime-specific innovation variances, and regime-conditional return distributions are consistent with financial intuition for a low-volatility mean-reverting regime and a high-volatility drifting regime (Cappé et al. 2005).

The central contribution is not a claim that HMMs universally outperform simpler models, but a careful, costed, walk-forward test of when regime awareness helps, when it fails, and how mathematical models behave when confronted with noisy, costly, non-stationary tick data (Cont 2001; Hautsch 2012).

Limitations. The exercise is restricted to three pair constructions, three independent one-month sub-samples (August 2024, September 2024, and August 2025), $K = 2$ regimes, unit-size long/flat/short positions, and a half-spread-plus-commission cost model that omits latency and market impact.

Future directions. The most natural extension is to combine the regime posteriors with a different entry signal in the drifting regime—for example, a momentum signal as in Avellaneda and Lee (2010)—rather than using the HMM only as a gate on a mean-reversion signal. A second direction is to pre-screen the pair universe with the baseline back-test before applying the HMM, avoiding modelling effort on pairs that the microstructure does not support (Krauss 2017).

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8 Appendix